

Course

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Essentials of Data Science Laboratory - 2304102L - 2026

Search course ctrl + k

3. Practical 3

3.1. Practice Lab Assignment

3.1.1. Numpy Array Operations

3.2. Lab Assignment

3.2.1. Numpy: Matrix Operations

3.2.2. Numpy: Horizontal and Vertical Stac...

3.2.3. Numpy: Custom Sequence Generation

3.2.4. Numpy: Arithmetic and Statistical Op...

3.2.5. Numpy: Copying and Viewing Arrays

3.2.6. Numpy: Searching, Sorting, Countin...

3.2.7. Student Data Analysis and Operations

4. Practical 4

4.1. Practice Lab Assignment

4.1.1. Pandas - series creation and manipu...

4.1.2. Dictionary to dataframe

4.1.3. Student Information

4.2. Lab Assignment

5. Practical 5

4.1.1. Pandas - series creation and manipulation

01:01

Write a Python program that takes a list of numbers from the user, creates a Pandas series from it, and then calculates the mean of even and odd numbers separately using the `groupby` and `mean()` operations.

Hint: You can group the Series based on whether each number is even or odd.

Input Format:

The user should enter a list of numbers separated by space when prompted.

Output Format:

The program should display the mean of even and odd numbers separately.

Each mean value should be displayed with a label indicating whether it corresponds to even or odd numbers.

Refer to the sample test cases for better understanding regarding input and output format.

Sample Test Cases

seriesMa...

Submit

1 import pandas as pd

2

3 # Take inputs from the user to create a list of numbers

4 numbers = list(map(int, input().split()))

5

6 # Create a Pandas series from the list of numbers

7 s = pd.Series(numbers)

8 # Grouping by even and odd numbers and calculating the mean

9 grouped = s.groupby(s % 2 == 0).mean()

10

11

12 # Display the mean of even and odd numbers with labels

13 grouped.index = ['Even' if is_even else 'Odd' for is_even in grouped.index]

14 print("Mean of even and odd numbers:")

15 print(grouped)

16

Average time 0.005 s 5.00 ms

Maximum time 0.013 s 13.00 ms

3 out of 3 shown test case(s) passed

3 out of 3 hidden test case(s) passed

Test case 1 13 ms Debug

Expected output

1 2 3 4 5 6 7 8 9 10

Mean of even and odd numbers:

Odd 5.0

Even ... 6.0

dtype: float64

Actual output

1 2 3 4 5 6 7 8 9 10

Mean of even and odd numbers:

Odd 5.0

Even ... 6.0

dtype: float64

Terminal Test cases

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09:47 07-05-2026

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Laboratory - 2304102L - 2026

Search coursectrl + k

3.2.2. Captain of the team

3. Practical 3

3.1. Practice Lab Assignment

3.1.1. Numpy Array Operations

3.2. Lab Assignment

3.2.1. Numpy: Matrix Operations

3.2.2. Numpy: Horizontal and Vertical Stac...

3.2.3. Numpy: Custom Sequence Generation

3.2.4. Numpy: Arithmetic and Statistical Op...

3.2.5. Numpy: Copying and Viewing Arrays

3.2.6. Numpy: Searching, Sorting, Countin...

3.2.7. Student Data Analysis and Operations

4. Practical 4

4.1. Practice Lab Assignment

4.1.1. Pandas - series creation and manipu...

4.1.2. Dictionary to dataframe

4.1.3. Student Information

4.2. Lab Assignment

5. Practical 5

4.1.2. Dictionary to dataframe00:40

A dictionary of lists has been provided to you in the editor. Create a DataFrame from the dictionary of lists and perform the listed operations, then display the DataFrame before and after each manipulation.

Create the DataFrame:

- Convert the dictionary to a Pandas DataFrame.

Add a new row:

- Take inputs from the user for the new row data (name, age).
- Add the new row to the DataFrame.
- Display the DataFrame after adding the new row.

Modify a row:

- Modify a specific row by changing the age. Take the row index and new age value from the user.
- Display the DataFrame after modifying the row.

Delete a row:

- Take the row index to be deleted from the user.
- Remove the specified row.
- Display the DataFrame after deleting the row.

Add a new column:

- Add a column **Gender** with values taken from the user.
- Display the DataFrame after adding the new column.

Modify a column:

- Convert names to uppercase.
- Display the DataFrame after modifying the column.

Delete a column:

- Remove the **Age** column.
- Display the DataFrame after deleting the column.

Sample Test Cases

datafram...

1'''import pandas as pd

2

3# Provided dictionary of lists

4data = {

5 'Name': ['Alice', 'Bob', 'Charlie'],

6 'Age': [25, 30, 35],

7}

8

9# Convert the dictionary to a DataFrame

10df = pd.DataFrame(data)

11

12# Display the original DataFrame

13print("Original DataFrame:")

14print(df)

15

16# Adding a new row

17

18

19# Display the DataFrame after adding a new row

20print("After adding a row:\n",df)

21

Average time0.037 s37.00 ms

Maximum time0.043 s43.00 ms

1 out of 1 shown test case(s) passed

1 out of 1 hidden test case(s) passed

Test case 143 msDebug

Expected output

Original DataFrame:

.....Name..Age

0.....Alice...25

1.....Bob...30

2.....Charlie...35

New name: Susan

Actual output

Original DataFrame:

.....Name..Age

0.....Alice...25

1.....Bob...30

2.....Charlie...35

New name: Susan

TerminalTest cases

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Search course

3.2.2. Captain of the team

3. Practical 3

3.1. Practice Lab Assignment

3.1.1. Numpy Array Operations

3.2. Lab Assignment

3.2.1. Numpy: Matrix Operations

3.2.2. Numpy: Horizontal and Vertical Stac...

3.2.3. Numpy: Custom Sequence Generation

3.2.4. Numpy: Arithmetic and Statistical Op...

3.2.5. Numpy: Copying and Viewing Arrays

3.2.6. Numpy: Searching, Sorting, Countin...

3.2.7. Student Data Analysis and Operations

4. Practical 4

4.1. Practice Lab Assignment

4.1.1. Pandas - series creation and manipu...

4.1.2. Dictionary to dataframe

4.1.3. Student Information

4.2. Lab Assignment

5. Practical 5

4.1.3. Student Information

00:16

Write a program to read a text file containing student information (name, age, and grade) using Pandas. Perform the following tasks:

• Display the first five rows of the data frame.

• Calculate the average age of the students (limit the average age up to 2 decimal places).

• Filter out the students who have a grade above a certain threshold (consider the threshold grade is 'B').

Note:
Refer to the displayed test cases for better understanding.

Sample Test Cases

studentin...

Submit

1 import pandas as pd

2

3 # Read the text file into a DataFrame

4 file = input()

5 data = pd.read_csv(file, sep="\s+", header=None, names=["Name", "Age", "Grade"])

6

7

8 # write your code here..

9 print("First five rows:")

10 print(data.head())

11

12 avg_age = round(data["Age"].mean(),2)

13 print(f"Average age: {avg_age}")

14

15 print("Students with a grade up to B")

16

17 filtered_student = data[data["Grade"] <= "B"]

18 print(filtered_student)

19

Average time
0.019 s
19.00 ms

Maximum time
0.028 s
28.00 ms

1 out of 1 shown test case(s) passed

1 out of 1 hidden test case(s) passed

Test case 1 28 ms Debug

Expected output
studentdata.txt

Actual output
studentdata.txt

First five rows:
Name Age Grade
0 John 25 A
1 Alin 22 B
2 Emma 24 A

Terminal Test cases

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Search course

3.2.4. Numpy: Arithmetic and Statistical Op...
3.2.5. Numpy: Copying and Viewing Arrays
3.2.6. Numpy: Searching, Sorting, Countin...
3.2.7. Student Data Analysis and Operations

4. Practical 4

4.1. Practice Lab Assignment

4.1.1. Pandas - series creation and manipu...
4.1.2. Dictionary to dataframe
4.1.3. Student Information

4.2. Lab Assignment

4.2.1. Month with the Highest Total Sales
4.2.2. Best Selling Product
4.2.3. City that Sold the Most Products
4.2.4. Most Frequently Sold Product Pairs
4.2.5. Titanic Dataset Analysis and Data Cl...
4.2.6. Titanic Dataset Analysis and Data Cl...
4.2.7. Titanic Dataset Analysis and Data Cl...
4.2.8. Titanic Dataset Analysis and Data Cl...

5. Practical 5

4.2.1. Month with the Highest Total Sales

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by Month and calculate the total sales for each month.
- Find the month with the highest total sales and display it.
- Also, display the total sales for the best month.

Sample Data:

```
Date,Product,Quantity,Price,City
2025-01-01,Product A,5,20,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,8,20,Los Angeles
2025-01-04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago
2025-01-05,Product C,10,30,Los Angeles
```

Note:
The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Sample Test Cases

monthFor...

```
1 # import pandas as pd
2
3 ## Prompt the user for the file name
4 # file_name = input()
5
6 ## Load the data
7 # df = pd.read_csv(file_name)
8
9
10 ## Find the month with the highest total sales
11 # best_month =
12 # highest_sales =
13
14 # print(f"Best month: {best_month}")
15 # print(f"Total sales: ${highest_sales:.2f}")
16 ## import pandas as pd
17
18 ## Prompt the user for the file name
19 # # file_name = input()
20
```

Average time: 0.010 s
Maximum time: 0.021 s

10.00 ms 21.00 ms

1 out of 1 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1 21 ms Debug

Expected output	Actual output
sales_data.csv	sales_data.csv
Best month: 2025-01	Best month: 2025-01
Total sales: \$1210.00	Total sales: \$1210.00

Terminal Test cases

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3.2.5. Numpy: Copying and Viewing Arrays

3.2.6. Numpy: Searching, Sorting, Countin...

3.2.7. Student Data Analysis and Operations

4. Practical 4

4.1. Practice Lab Assignment

4.1.1. Pandas - series creation and manipu...

4.1.2. Dictionary to dataframe

4.1.3. Student Information

4.2. Lab Assignment

4.2.1. Month with the Highest Total Sales

4.2.2. Best Selling Product

4.2.3. City that Sold the Most Products

4.2.4. Most Frequently Sold Product Pairs

4.2.5. Titanic Dataset Analysis and Data Cl...

4.2.6. Titanic Dataset Analysis and Data Cl...

4.2.7. Titanic Dataset Analysis and Data Cl...

4.2.8. Titanic Dataset Analysis and Data Cl...

5. Practical 5

4.2.2. Best Selling Product

00:31

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

The CSV file contains the columns: Date, Product, Quantity, Price, and City.

Find the product that sold the most in terms of quantity sold.

Display the product that sold the most and the total quantity sold for that product.

Sample Data:

Date	Product	Quantity	Price	City
2025-01-01	Product A	5	20	New York
2025-01-01	Product B	3	15	Los Angeles
2025-01-02	Product A	7	20	New York
2025-01-02	Product C	4	30	Chicago
2025-01-03	Product B	2	15	Chicago
2025-01-03	Product A	8	20	Los Angeles
2025-01-04	Product C	6	30	New York
2025-01-04	Product B	5	15	Los Angeles
2025-01-05	Product A	3	20	Chicago
2025-01-05	Product C	10	30	Los Angeles

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Sample Test Cases

monthFor...

Submit

1

import pandas as pd

2

3

Prompt the user for the file name

4

file_name = input()

5

6

Load the data

7

df = pd.read_csv(file_name)

8

9

10

product_sales=df.groupby('Product')['Quantity'].sum()

11

Find the product with the highest total quantity sold

12

best_product =product_sales.idxmax()

13

highest_quantity = product_sales.max()

14

15

Display the result

16

print(f"Best selling product: {best_product}")

17

print(f"Total quantity sold: {highest_quantity}")

18

Average time

0.010 s

10.00 ms

Maximum time

0.019 s

19.00 ms

1 out of 1 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

19 ms

Debug

Expected output

Actual output

sales_data.csv

sales_data.csv

Best-selling product: Product A

Best-selling product: Product A

Total quantity sold: 23

Total quantity sold: 23

Terminal

Test cases

Prev

Reset

Submit

Next

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09:47

07-05-2026

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3.2.4. Numpy: Arithmetic and Statistical Op...

3.2.5. Numpy: Copying and Viewing Arrays

3.2.6. Numpy: Searching, Sorting, Countin...

3.2.7. Student Data Analysis and Operations

4. Practical 4

4.1. Practice Lab Assignment

4.1.1. Pandas - series creation and manipu...

4.1.2. Dictionary to dataframe

4.1.3. Student Information

4.2. Lab Assignment

4.2.1. Month with the Highest Total Sales

4.2.2. Best Selling Product

4.2.3. City that Sold the Most Products

4.2.4. Most Frequently Sold Product Pairs

4.2.5. Titanic Dataset Analysis and Data Cl...

4.2.6. Titanic Dataset Analysis and Data Cl...

4.2.7. Titanic Dataset Analysis and Data Cl...

4.2.8. Titanic Dataset Analysis and Data Cl...

5. Practical 5

4.2.3. City that Sold the Most Products

00:17

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

The CSV file contains the columns: Date, Product, Quantity, Price, and City.

Group the data by city and calculate the total quantity of products sold for each city.

Find the city that sold the most products (based on the total quantity sold).

Sample Data:

Date	Product	Quantity	Price	City
2025-01-01	Product A	5	20	New York
2025-01-01	Product B	3	15	Los Angeles
2025-01-02	Product A	7	20	New York
2025-01-02	Product C	4	30	Chicago
2025-01-03	Product B	2	15	Chicago
2025-01-03	Product A	8	20	Los Angeles
2025-01-04	Product C	6	30	New York
2025-01-04	Product B	5	15	Los Angeles
2025-01-05	Product A	3	20	Chicago
2025-01-05	Product C	10	30	Los Angeles

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Sample Test Cases

monthFor...

Submit

1

import pandas as pd

2

3

Prompt the user for the file name

4

file_name = input()

5

6

Load the data

7

df = pd.read_csv(file_name)

8

9

write the code..

10

product_sales=df.groupby('City')['Quantity'].sum()

11

best_city = product_sales.idxmax()

12

13

14

Display the result

15

print(f"City sold the most products: {best_city}")

16

Average time

0.009 s

9.33 ms

Maximum time

0.019 s

19.00 ms

1 out of 1 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

19 ms

Debug

Expected output

sales_data.csv

City sold the most products: Los Angeles

Actual output

sales_data.csv

City sold the most products: Los Angeles

Terminal

Test cases

30°C

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Search

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07-05-2026

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3.2.4. Numpy: Arithmetic and Statistical Op...
3.2.5. Numpy: Copying and Viewing Arrays
3.2.6. Numpy: Searching, Sorting, Countin...
3.2.7. Student Data Analysis and Operations

4. Practical 4

4.1. Practice Lab Assignment
4.1.1. Pandas - series creation and manipu...
4.1.2. Dictionary to dataframe
4.1.3. Student Information

4.2. Lab Assignment
4.2.1. Month with the Highest Total Sales
4.2.2. Best Selling Product
4.2.3. City that Sold the Most Products
4.2.4. Most Frequently Sold Product Pairs
4.2.5. Titanic Dataset Analysis and Data CL...
4.2.6. Titanic Dataset Analysis and Data CL...
4.2.7. Titanic Dataset Analysis and Data CL...
4.2.8. Titanic Dataset Analysis and Data CL...

5. Practical 5

4.2.4. Most Frequently Sold Product Pairs

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the following columns: Date, Product, Quantity, Price, and City.
- For each date, find all pairs of products that were sold together (i.e., two products sold on the same date).
- Output the product pair/s that was sold most frequently.

Sample Data:

Date	Product	Quantity	Price	City
2025-01-01	Product A	5	20	New York
2025-01-01	Product B	3	15	Los Angeles
2025-01-02	Product A	7	20	New York
2025-01-02	Product C	4	30	Chicago
2025-01-03	Product B	2	15	Chicago
2025-01-03	Product A	8	20	Los Angeles
2025-01-04	Product C	6	30	New York
2025-01-04	Product B	5	15	Los Angeles
2025-01-05	Product A	3	20	Chicago
2025-01-05	Product C	10	30	Los Angeles

Explanation:
Transactions:

- 2025-01-01: Product A, Product B
- 2025-01-02: Product A, Product C
- 2025-01-03: Product B, Product A
- 2025-01-04: Product C, Product B
- 2025-01-05: Product A, Product C

Now, let's count how often the pairs of products appear together:

- Product A and Product B: Appear in transactions on 2025-01-01 and 2025-01-03.
- Product A and Product C: Appear in transactions on 2025-01-02 and 2025-01-05.
- Product B and Product C: Appears in transactions on 2025-01-04.

Most Frequent Product Combinations:

- Product A and Product B (2 times)
- Product A and Product C (2 times)

Note:

frequenti...

```
1 # import pandas as pd
2 # from itertools import combinations
3 # from collections import Counter
4
5 ## Prompt user to input the file name
6 # file_name = input()
7
8 ## Read data from the specified CSV file
9 # df = pd.read_csv(file_name)
10
11 ## # write the code
12
13
14 ## # Output the most frequent product pairs
15 ## import pandas as pd
16 ## from itertools import combinations
17 ## from collections import Counter
18
19 ## # Prompt user to input the file name
20 # # file_name = input()
```

Average time
0.010 s
9.87 ms

Maximum time
0.020 s
20.00 ms

1 out of 1 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1 20 ms Debug

Expected output
sales_data.csv
Product A and Product B: 2 times
Product A and Product C: 2 times

Actual output
sales_data.csv
Product A and Product B: 2 times
Product A and Product C: 2 times

Terminal Test cases

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Laboratory - 2304102L - 2026

Search course

3.2.4. Numpy: Arithmetic and Statistical Op...

3.2.5. Numpy: Copying and Viewing Arrays

3.2.6. Numpy: Searching, Sorting, Countin...

3.2.7. Student Data Analysis and Operations

4. Practical 4

4.1. Practice Lab Assignment

4.1.1. Pandas - series creation and manipu...

4.1.2. Dictionary to dataframe

4.1.3. Student Information

4.2. Lab Assignment

4.2.1. Month with the Highest Total Sales

4.2.2. Best Selling Product

4.2.3. City that Sold the Most Products

4.2.4. Most Frequently Sold Product Pairs

4.2.5. Titanic Dataset Analysis and Data Cl...

4.2.6. Titanic Dataset Analysis and Data Cl...

4.2.7. Titanic Dataset Analysis and Data Cl...

4.2.8. Titanic Dataset Analysis and Data Cl...

5. Practical 5

4.2.5. Titanic Dataset Analysis and Data Cleaning

00:27

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset. For each question, perform necessary data cleaning, transformations, and calculations as required.

1. Display the first 5 rows of the dataset.

2. Display the last 5 rows of the dataset.

3. Get the shape of the dataset (number of rows and columns).

4. Get a summary of the dataset (using `.info()`).

5. Get basic statistics (mean, standard deviation, etc.) of the dataset using `.describe()`.

6. Check for missing values and display the count of missing values for each column.

7. Fill missing values in the 'Age' column with the median age.

8. Fill missing values in the 'Embarked' column with the most frequent value (`mode()`).

9. Drop the 'Cabin' column due to many missing values.

10. Create a new column, 'FamilySize' by adding the 'SibSp' and 'Parch' columns.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S

9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S

titanicDat...

1 # import pandas as pd

2 # import numpy as np

3

4 ## Load the Titanic dataset

5 # data = pd.read_csv('Titanic-Dataset.csv')

6

7 ## 1. Display the first 5 rows of the dataset

8

9

10 ## 2. Display the last 5 rows of the dataset

11

12

13 ## 3. Get the shape of the dataset

14

15

16 ## 4. Get a summary of the dataset (info)

17

18

19 ## 5. Get basic statistics of the dataset

20

Average time 0.074 s 74.00 ms

Maximum time 0.074 s 74.00 ms

1 out of 1 shown test case(s) passed

Test case 1 74 ms

Expected output

Actual output

Terminal

Test cases

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Search

ENG IN

09:49 07-05-2026

Essentials of Data Science Laboratory - 2304102L - 2026

Search course

3.2.4. Numpy: Arithmetic and Statistical Op...

3.2.5. Numpy: Copying and Viewing Arrays

3.2.6. Numpy: Searching, Sorting, Countin...

3.2.7. Student Data Analysis and Operations

4. Practical 4

4.1. Practice Lab Assignment

4.1.1. Pandas - series creation and manipu...

4.1.2. Dictionary to dataframe

4.1.3. Student Information

4.2. Lab Assignment

4.2.1. Month with the Highest Total Sales

4.2.2. Best Selling Product

4.2.3. City that Sold the Most Products

4.2.4. Most Frequently Sold Product Pairs

4.2.5. Titanic Dataset Analysis and Data Cl...

4.2.6. Titanic Dataset Analysis and Data Cl...

4.2.7. Titanic Dataset Analysis and Data Cl...

4.2.8. Titanic Dataset Analysis and Data Cl...

5. Practical 5

4.2.6. Titanic Dataset Analysis and Data Cleaning - 2

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Create a new column 'IsAlone' which is 1 if the passenger is alone (FamilySize = 0), otherwise 0.

2. Convert the 'Sex' column to numeric values (male: 0, female: 1).

3. One-hot encode the 'Embarked' column, dropping the first category.

4. Get the mean age of passengers.

5. Get the median fare of passengers.

6. Get the number of passengers by class.

7. Get the number of passengers by gender.

8. Get the number of passengers by survival status.

9. Calculate the survival rate of passengers.

10. Calculate the survival rate by gender.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nesser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

titanicDat...

```
1 # import pandas as pd
2 # import numpy as np
3
4 ## Load the Titanic dataset
5 # data = pd.read_csv('Titanic-Dataset.csv')
6 # data['FamilySize'] = data['SibSp'] + data['Parch']
7
8 ## 1. Create a new column 'IsAlone' (1 if alone, 0 otherwise)
9
10
11 ## 2. Convert 'Sex' to numeric (male: 0, female: 1)
12
13
14 ## 3. One-hot encode the 'Embarked' column
15
16
17 ## 4. Get the mean age of passengers
18
19
20
```

Average time0.031 s31.00 ms

Maximum time0.031 s31.00 ms

1 out of 1 shown test case(s) passed

Test case 131 ms

Debug

Expected output	Actual output
29.69911764705882	29.69911764705882
14.4542	14.4542
3...491	3...491
1...216	1...216
2...184	2...184
Name: Pclass, dtype: int64	Name: Pclass, dtype: int64

Terminal

Test cases

4.2.7. Titanic Dataset Analysis and Data Cleaning - 3

00:20

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Calculate the survival rate by class.
2. Calculate the survival rate by embarkation location (Embarked_S).
3. Calculate the survival rate by family size (FamilySize).
4. Calculate the survival rate by being alone (IsAlone).
5. Get the average fare by passenger class (Pclass).
6. Get the average age by passenger class (Pclass).
7. Get the average age by survival status (Survived).
8. Get the average fare by survival status (Survived).
9. Get the number of survivors by class (Pclass).
10. Get the number of non-survivors by class (Pclass).

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nesser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C
```

titanicDat...

Submit

1 # import pandas as pd

2 # import numpy as np

3

4 ## Load the Titanic dataset

5 # data = pd.read_csv('Titanic-Dataset.csv')

6 # data['FamilySize'] = data['SibSp'] + data['Parch']

7 # data['IsAlone'] = np.where(data['FamilySize'] > 0, 0, 1)

8 # data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

9

10 ## 1. Calculate the survival rate by class

11

12

13 ## 2. Calculate the survival rate by embarked location

14

15

16 ## 3. Calculate the survival rate by family size

17

18

19 ## 4. Calculate the survival rate by being alone

20

Average time

0.045 s

45.00 ms

Maximum time

0.045 s

45.00 ms

1 out of 1 shown test case(s) passed

Test case 1 45 ms Debug

Expected output

Actual output

Pclass

Pclass

1...0.629630

1...0.629630

2...0.472826

2...0.472826

3...0.242363

3...0.242363

Name: Survived, dtype: float64

Name: Survived, dtype: float64

Embarked_S

Embarked_S

Terminal

Test cases

Pin

ctrl + k

5. Practical 5
-

00:09 AA [icon] [icon] -

1. Get the number of survivors by gender (Sex).
2. Get the number of non-survivors by gender (Sex).
3. Get the number of survivors by embarkation location (Embarked_S).
4. Get the number of non-survivors by embarkation location (Embarked_S).
5. Calculate the percentage of children (Age < 18) who survived.
6. Calculate the percentage of adults (Age >= 18) who survived.
7. Get the median age of survivors.
8. Get the median age of non-survivors.
9. Get the median fare of survivors.
10. Get the median fare of non-survivors.

[illegible]

```

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,ParCh,Ticket,Fare,Cabin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2 3101282,7.92,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,0,0,330877,8.4583,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,S
10,1,2,"Nasser, Mrs. Nicholas (Adèle Achém)",female,14,1,0,237736,30.0708,C

```

Test case 1 34 ms Debug	
Expected output	Actual output
female ... 233	female ... 233
male ... 109	male ... 109
Name: Sex, dtype: int64	Name: Sex, dtype: int64
male ... 468	male ... 468
female ... 81	female ... 81
Name: Sex, dtype: int64	Name: Sex, dtype: int64